

CONCEPT 31.4

Fungi have radiated into a diverse set of lineages

(pp. 660–667)

Fungal Phylum	Distinguishing Features
Cryptomycota (cryptomycetes)	Parasites with flagellated spores 
Microsporidia (microsporidians)	Parasitic cells that form resistant spores 
Chytridiomycota (chytrids)	Flagellated spores 
Zoopagomycota (zoopagomycetes)	Resistant zygosporangium as sexual stage 
Mucromycota (mucromycetes)	Include fungi that form arbuscular mycorrhizae with plants 
Ascomycota (ascomycetes)	Sexual spores (ascospores) borne internally in sacs called asci; vast numbers of asexual spores (conidia) produced 
Basidiomycota (basidiomycetes)	Elaborate fruiting body (basidiocarp) containing many basidia that produce sexual spores (basidiospores) 

DRAW IT Draw a phylogenetic tree of the major groups of fungi.

CONCEPT 31.5

Fungi play key roles in nutrient cycling, ecological interactions, and human welfare (pp. 667–671)

- Fungi perform essential recycling of chemical elements between the living and nonliving world.
- Lichens** are highly integrated symbiotic associations of fungi and algae or cyanobacteria.
- Many fungi are parasites, mostly of plants.
- Humans use fungi for food and to make antibiotics.

? How are fungi important as decomposers, mutualists, and pathogens?

TEST YOUR UNDERSTANDING

➔ For more multiple-choice questions, go to the **Practice Test** in the **Mastering Biology** eText or Study Area, or go to goo.gl/GruWRg.

Levels 1-2: Remembering/Understanding

- All fungi are
 - symbiotic.
 - heterotrophic.
 - flagellated.
 - decomposers.
- Which of the following cells or structures are associated with asexual reproduction in fungi?
 - ascospores
 - basidiospores
 - zygosporangia
 - conidiophores
- The closest relatives of fungi are thought to be the
 - animals.
 - vascular plants.
 - mosses.
 - slime molds.

Levels 3-4: Applying/Analyzing

- The most important adaptive advantage associated with the filamentous nature of fungal mycelia is
 - the ability to form haustoria and parasitize other organisms.
 - the potential to inhabit almost all terrestrial habitats.
 - the increased chance of contact between mating types.
 - an extensive surface area well suited for invasive growth and absorptive nutrition.
- SCIENTIFIC INQUIRY • INTERPRET THE DATA** The grass *Dichanthelium lanuginosum* lives in hot soils and houses fungi of the genus *Curvularia* as endophytes. Researchers tested the impact of *Curvularia* on the heat tolerance of this grass. They grew plants without (E–) and with (E+) *Curvularia* endophytes at different temperatures and measured plant mass and the number of new shoots the plants produced. Draw a bar graph for plant mass versus temperature and interpret it.

Soil Temp.	<i>Curvularia</i> + or –	Plant Mass (g)	No. of New Shoots
30°C	E–	16.2	32
	E+	22.8	60
35°C	E–	21.7	43
	E+	28.4	60
40°C	E–	8.8	10
	E+	22.2	37
45°C	E–	0	0
	E+	15.1	24

Data from R. S. Redman et al., Thermotolerance generated by plant/fungal symbiosis, *Science* 298:1581 (2002).

Levels 5-6: Evaluating/Creating

- EVOLUTION CONNECTION** The fungus-alga symbiosis that makes up a lichen is thought to have evolved multiple times independently in different fungal groups. However, lichens fall into three well-defined growth forms (see Figure 31.24). How could you test the following hypotheses? Hypothesis 1: Crustose, foliose, and fruticose lichens each represent a monophyletic group. Hypothesis 2: Each lichen growth form represents convergent evolution by diverse fungal groups.
- WRITE ABOUT A THEME: ORGANIZATION** As you read in this chapter, fungi have long formed symbiotic associations with plants and with algae. In a short essay (100–150 words), describe how these two types of associations may lead to emergent properties in biological communities.
- SYNTHESIZE YOUR KNOWLEDGE**



This wasp is the unfortunate victim of an entomopathogenic fungus (a parasitic fungus of insects). Write a paragraph describing what this image illustrates about the nutritional mode, body structure, and ecological role of the fungus.

For selected answers, see Appendix A.